

Wavin AquaCell PT-ROW™ Sizing and Design Guidance

Introduction

The Wavin AquaCell PT-ROW™ (Pre-Treatment Row) is a modular component of the AquaCell underground stormwater management system. It is designed to capture sediment, debris, and pollutants before water enters the main storage volume. Proper sizing of the PT-ROW™ is essential to ensure effective treatment, reduce maintenance, and comply with local stormwater regulations. This document will aid in the design and implementation of a PT-ROW™ within the AquaCell stormwater system.

Key Design Questions & Assumptions

- What is the required pre-treatment volume?
- What is the required treatment flow rate for the pre-treatment system?
- What are the dimensions and void space percentage of the AquaCell crates?
 - AquaCell void ratio: 95% (typical)
 - Flow is horizontal through the PT-ROW™
 - Each crate has a footprint of 1.9685 ft x 3.937 ft $\approx$ 7.75 ft²
- What are the required pollutant removal targets for the pre-treatment system?
 - Sediment loading based on local design storm and catchment characteristics
- Are there any site-specific constraints (footprint limitations, soil conditions)?
- Are there any flow diversion structures (OCS, bypasses, etc.) that are part of the overall system design?

Volume-Based Sizing Method

This method is commonly used when the primary goal is to capture and hold a specific volume of runoff for settling and filtration. The number of crates required is calculated using the following formula:

$$\text{Number of Crates} = \text{Required Pre-treatment Volume} / \text{Volume of a Single Crate}$$

Where:

- Required Pre-treatment Volume is provided by the engineer.
- Volume of a Single Crate is calculated using crate dimensions and void space percentage.

Flow-Based Sizing Method

This method is used when the pre-treatment system must handle a specific treatment flow rate. It is particularly relevant for systems with flow control or bypass structures.

$$\text{Number of Crates} = (\text{Q}) \text{ treatment} \times \text{Coefficient}$$

Where:

- (Q) treatment is the treatment flow rate (CFS).
- Coefficient is based on crate design and performance (0.464 CFS per AquaCell module).

Combined Approach

In some cases, both volume and flow criteria are used to size the PT-ROW™. The final design should be reviewed by a qualified engineer to ensure compliance with local regulations and site-specific needs.

Sample Calculations

1. Volume-Based Example:

- Required Pre-treatment Volume = 200 ft³
- Volume of a Single Crate = 10.25 ft³
- Number of Crates = 200 / 10.25 ≈ 19.51 → Round up to 20 crates

2. Flow-Based Example:

- (Q) treatment = 2.5 CFS
- Flow capacity per AquaCell module = 0.464 CFS/module
- Number of Crates = 2.5 / 0.464 ≈ 5.39 → Round up to 6 crates

Final Recommendations

Always consult with a qualified civil engineer to determine the appropriate sizing method and confirm all design assumptions. Use manufacturer-provided tools and data where available. Consider both hydraulic and pollutant removal performance and ensure the PT-ROW™ is accessible for inspection and maintenance.

Recommended woven fabric for PT-ROW™ installation

Woven Monofilament Geotextile Specification (Generic Callout)

Provide a woven monofilament geotextile composed of high-tenacity polypropylene yarns, designed for separation, filtration, and erosion protection. The geotextile shall meet the following minimum properties:

- Grab Tensile Strength: ≥ 365 lbs (MD), ≥ 200 lbs (CD) per ASTM D4632
- Trapezoid Tear Strength: ≥ 115 lbs (MD), ≥ 75 lbs (CD) per ASTM D4533
- CBR Puncture Strength: ≥ 675 lbs per ASTM D6241
- Permittivity: $\geq 2.1 \text{ sec}^{-1}$ per ASTM D4491
- Flow Rate: $\geq 145 \text{ gal/min/ft}^2$ per ASTM D4491
- Apparent Opening Size (AOS): $\leq$ No. 40 sieve (0.425 mm) per ASTM D4751
- UV Resistance: $\geq 90\%$ retained strength after 500 hours per ASTM D4355
- Percent Open Area: $\geq 10\%$ per COE-02215

Geotextile shall be inert to biological degradation and resistant to naturally encountered chemicals, alkalis, and acids. Roll dimensions shall be approximately 12.5 ft x 300 ft.